VyOS – Fullständig Konfigurationsfil (Alla kommandon i ordning)

Denna pdf innehåller **alla kommandon** som använts för att konfigurera nätverket, DHCP, DNS, NAT, brandvägg och VG-förbättringar.

1. Grundläggande Setup

1.1 Tangentbordslayout

```
configure
set system option keyboard-layout se-latin1
commit
save
exit
```

2. Interface-konfiguration

2.1 WAN (eth0)

```
configure
set interfaces ethernet eth0 address dhcp
set interfaces ethernet eth0 description 'WAN'
```

2.2 LAN1 (eth1)

```
set interfaces ethernet eth1 address 192.168.0.1/24
set interfaces ethernet eth1 description 'LAN1'
```

2.3 LAN2 (eth2)

```
set interfaces ethernet eth2 address 192.168.1.1/24
set interfaces ethernet eth2 description 'LAN2'
commit
save
exit
```

3. DHCP-server

3.1 LAN1 – 192.168.0.0/24

```
configure
set service dhcp-server shared-network-name LAN1 subnet 192.168.0.0/24 option
default-router 192.168.0.1
set service dhcp-server shared-network-name LAN1 subnet 192.168.0.0/24 option
name-server 192.168.0.1
set service dhcp-server shared-network-name LAN1 subnet 192.168.0.0/24 lease 86400
set service dhcp-server shared-network-name LAN1 subnet 192.168.0.0/24 range 0 start
```

192.168.0.20
set service dhcp-server shared-network-name LAN1 subnet 192.168.0.0/24 range 0 stop 192.168.0.50
set service dhcp-server shared-network-name LAN1 subnet 192.168.0.0/24 subnet-id 1

3.2 LAN2 – 192.168.1.0/24

set service dhcp-server shared-network-name LAN2 subnet 192.168.1.0/24 option default-router 192.168.1.1
set service dhcp-server shared-network-name LAN2 subnet 192.168.1.0/24 option name-server 192.168.1.1
set service dhcp-server shared-network-name LAN2 subnet 192.168.1.0/24 lease 86400
set service dhcp-server shared-network-name LAN2 subnet 192.168.1.0/24 range 0 start 192.168.1.20
set service dhcp-server shared-network-name LAN2 subnet 192.168.1.0/24 range 0 stop 192.168.1.50
set service dhcp-server shared-network-name LAN2 subnet 192.168.1.0/24 subnet-id 2
commit
save
exit

4. DNS-Forwarding

configure
set service dns forwarding cache-size 0
set service dns forwarding listen-address 192.168.0.1
set service dns forwarding listen-address 192.168.1.1
set service dns forwarding allow-from 192.168.0.0/24
set service dns forwarding allow-from 192.168.1.0/24
commit
save
exit

5. NAT – Internetåtkomst

5.1 LAN1 NAT

configure
set nat source rule 100 outbound-interface name eth0
set nat source rule 100 source address 192.168.0.0/24
set nat source rule 100 translation address masquerade

5.2 LAN2 NAT

set nat source rule 110 outbound-interface name eth0
set nat source rule 110 source address 192.168.1.0/24
set nat source rule 110 translation address masquerade
commit

save
exit

6. Avancerad Brandvägg

6.1 Common Chain – Connection Tracking

configure
set firewall ipv4 name CONN_FILTER default-action 'return'

set firewall ipv4 name CONN_FILTER rule 10 action 'accept'
set firewall ipv4 name CONN_FILTER rule 10 state established 'enable'
set firewall ipv4 name CONN_FILTER rule 10 state related 'enable'

set firewall ipv4 name CONN_FILTER rule 20 action 'drop'
set firewall ipv4 name CONN_FILTER rule 20 state invalid 'enable'

6.2 WAN_LOCAL – Trafik från WAN till routern

set firewall ipv4 name WAN_LOCAL default-action 'drop'

set firewall ipv4 name WAN_LOCAL rule 10 action 'accept'
set firewall ipv4 name WAN_LOCAL rule 10 protocol 'icmp'

6.3 WAN_IN – Trafik från WAN genom routern

set firewall ipv4 name WAN_IN default-action 'drop'

6.4 LAN_LOCAL – Trafik från LAN till routern

set firewall ipv4 name LAN_LOCAL default-action 'drop'

set firewall ipv4 name LAN_LOCAL rule 10 action 'accept'
set firewall ipv4 name LAN_LOCAL rule 10 protocol 'icmp'

6.5 LAN_IN – Trafik från LAN genom routern

set firewall ipv4 name LAN_IN default-action 'accept'

6.6 Bind brandväggsregler till interfaces

WAN (eth0)

set interfaces ethernet eth0 firewall ipv4 local 'WAN_LOCAL'
set interfaces ethernet eth0 firewall ipv4 forward 'WAN_IN'

LAN1 (eth1)

set interfaces ethernet eth1 firewall ipv4 local 'LAN_LOCAL'
set interfaces ethernet eth1 firewall ipv4 forward 'LAN_IN'

LAN2 (eth2)

```
set interfaces ethernet eth2 firewall ipv4 local 'LAN_LOCAL'
set interfaces ethernet eth2 firewall ipv4 forward 'LAN_IN'
```

6.7 Jump-rules för CONN_FILTER

```
set firewall ipv4 input filter rule 10 action 'jump'
set firewall ipv4 input filter rule 10 jump-target 'CONN_FILTER'

set firewall ipv4 forward filter rule 10 action 'jump'
set firewall ipv4 forward filter rule 10 jump-target 'CONN_FILTER'
commit
save
exit
```

7. VG-Förbättringar brandvägg

7.1 Rate-limit ICMP (skydd mot ICMP-flood)

```
configure
set firewall ipv4 name WAN_LOCAL rule 30 description 'Rate limit ICMP'
set firewall ipv4 name WAN_LOCAL rule 30 action 'accept'
set firewall ipv4 name WAN_LOCAL rule 30 protocol 'icmp'
set firewall ipv4 name WAN_LOCAL rule 30 limit rate '10/second'
```

7.2 Bogon-filter (skydd mot IP-spoofing)

```
set firewall ipv4 name WAN_IN rule 40 description 'Drop RFC1918 from WAN'
set firewall ipv4 name WAN_IN rule 40 action 'drop'
set firewall ipv4 name WAN_IN rule 40 source address '10.0.0.0/8'

set firewall ipv4 name WAN_IN rule 41 action 'drop'
set firewall ipv4 name WAN_IN rule 41 source address '172.16.0.0/12'

set firewall ipv4 name WAN_IN rule 42 action 'drop'
set firewall ipv4 name WAN_IN rule 42 source address '192.168.0.0/16'
```

7.3 Logging av droppad WAN-trafik

```
set firewall ipv4 name WAN_IN rule 5 description 'Log dropped WAN traffic'
set firewall ipv4 name WAN_IN rule 5 action 'drop'
set firewall ipv4 name WAN_IN rule 5 log 'enable'
commit
save
exit
```

8. VG-Förbättringar Monitoring & Event-Handling

8.1 Monitoring – Telegraf (systemövervakning)

```
configure
set service monitoring telegraf enable
set service monitoring telegraf output influxdb url 'http://192.168.0.10:8086'
commit
save
exit
```

8.2 Event-Handler – Interface Down Detection

```
configure
set service event-handler interface-down event 'interface eth0 down'
set service event-handler interface-down execute '/config/scripts/wan-down.sh'
commit
save
exit
```